

Core Scenario Table

ID	Env	Scenario	Intent	Emotion	Gesture	Motion	Context	Difficulty
C1	Classroom	Student with raised hand, confused	HELP_REQUEST	Sad	One hand raised	Stationary	Near desks	Easy
C2	Classroom	Student passing with quick wave	NEUTRAL_PASS	Happy	Brief wave	Fast, across	Corridor	Medium
C3	Classroom	Teacher pointing at door	GIVE_WAY	Angry	Pointing	Approaching	Doorway	Easy
C4	Classroom	Student carrying books	GIVE_WAY	Neutral	None	Slow approach	Near shelves	Easy
C5	Classroom	Panicking student running	EMERGENCY	Fearful	Arms waving	Fast toward robot	Open area	Medium
O1	Office	Worker passing with wave	NEUTRAL_PASS	Neutral	Brief wave	Fast, across	Corridor	Medium
O2	Office	Worker pointing at blocked shelf	GIVE_WAY	Angry	Pointing	Approaching	Shelf area	Easy
O3	Office	Colleague greeting	GREETING	Happy	Wave	Approach + stop	Open area	Easy
O4	Office	Sad worker beckoning	HELP_REQUEST	Sad	Beckoning	Stationary	Workstation	Easy
O5	Office	Collapsed person reaching	EMERGENCY	Fearful	Reaching	Minimal	Floor	Medium
K1	Kitchen	Frustrated cook beckoning	HELP_REQUEST	Angry	Beckoning	Stationary	Counter	Easy
K2	Kitchen	Person backing from stove	EMERGENCY	Surprised	Arms up	Backing away	Stove	Medium
K3	Kitchen	Cook pointing at cabinet	TASK_ASSIST	Neutral	Pointing	Stationary	Cabinet	Easy
K4	Kitchen	Cook carrying pot	GIVE_WAY	Neutral	None	Approaching	Narrow path	Easy
K5	Kitchen	Cook walking across	NEUTRAL_PASS	Happy	None	Across	Kitchen	Easy

Claims Table

ID	A (All cues)	B (Conflict)	C (Missing cue)	D (Baseline)
C1	✓	✓	–	–
C2	✓	✓	–	–
C3	✓	✓	–	–
C4	✓	–	✓	–
C5	–	–	–	✓
O1	✓	✓	–	–
O2	✓	✓	–	–
O3	–	–	–	✓
O4	✓	✓	–	–
O5	✓	–	✓	–
K1	✓	✓	–	–
K2	✓	✓	–	–
K3	✓	✓	–	–
K4	✓	–	✓	–
K5	–	✓	–	✓

Rule-Based Failure vs Fusion Outcome

ID	Rule-Based Output	Correct (Fusion)
C1	STOP / GREETING	HELP_REQUEST
C2	GREETING	NEUTRAL_PASS
C3	THREAT_AVOIDANCE	GIVE_WAY
C4	No action	GIVE_WAY
C5	EMERGENCY	EMERGENCY
O1	GREETING	NEUTRAL_PASS
O2	THREAT_AVOIDANCE	GIVE_WAY
O3	GREETING	GREETING
O4	Undefined / conflict	HELP_REQUEST
O5	No emergency detected	EMERGENCY
K1	Undefined / conflict	HELP_REQUEST
K2	Ambiguous / neutral	EMERGENCY
K3	Navigate only	TASK_ASSIST
K4	No action	GIVE_WAY
K5	GREETING	NEUTRAL_PASS

Scenario Purpose Table (Why each exists)

Type	Scenarios
Conflict (B)	C1, C2, C3, O1, O2, O4, K1, K2, K3, K5
Ablation (C)	C4, O5, K4
Baseline (D)	C5, O3, K5
Flagship	K1